FAROUK OSMAN ASSIGNMENTS

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ASSIGNMENT 1

##QN 1; Loading data sets

-Malaria cases, suspected, number.csv

-MALARIA_TOTAL_CASES.dta

-ITN Access.xlsx (Read both sheets)

```
#loading( Malaria cases, suspected, number.csv using read.csv)
malaria_cases_df <- read.csv(here::here("data", "raw", "Malaria cases, suspected, number.csv"))
```

loading MALARIA_TOTAL_CASES.dta

```
#loading MALARIA_TOTAL_CASES.dta using haven package
#haven package installed in console
library(haven)
malaria_total_cases_df <- read_dta(here::here("data", "raw", "MALARIA_TOTAL_CASES.dta"))
```

Loading ITN Access; both sheets

```
#Loading ITN Access using the readxl package
#readxl package installed in console

library(readxl)
itn_access1 <- read_excel(here::here("data", "raw", "ITN Access.xlsx"), sheet = 1)   # read sheet 1

itn_access2 <- read_excel(here::here("data", "raw", "ITN Access.xlsx"), sheet = 2)  #read sheet 2
```

QN 2; EXPLORING THE DATASETS

```
View(malaria_cases_df)  #view malaria_cases_df

head(malaria_total_cases_df, 10) #obtain first 10 rows of dataframe
```

```
## # A tibble: 10 x 25
##         id indicatorcode       spatialdimension spatialdimensionvaluecode
##      <dbl> <chr>               <chr>            <chr>
```

```
##  1 6653181 MALARIA_TOTAL_CASES COUNTRY          PAK
##  2 5140438 MALARIA_TOTAL_CASES COUNTRY          MWI
##  3 3504290 MALARIA_TOTAL_CASES COUNTRY          MDG
##  4 8663118 MALARIA_TOTAL_CASES COUNTRY          CAF
##  5 4520145 MALARIA_TOTAL_CASES COUNTRY          AFG
##  6 6939956 MALARIA_TOTAL_CASES COUNTRY          MLI
##  7 3804728 MALARIA_TOTAL_CASES COUNTRY          AFG
##  8 9255117 MALARIA_TOTAL_CASES COUNTRY          MOZ
##  9 2516740 MALARIA_TOTAL_CASES COUNTRY          ZAF
## 10 2556863 MALARIA_TOTAL_CASES COUNTRY          BRA
## # i 21 more variables: parentlocationcode <chr>, parentlocation <chr>,
## #   timedimension <chr>, timedim <dbl>, disaggregatingdimension1 <dbl>,
## #   disaggregatingdimension1valuecod <dbl>, disaggregatingdimension2 <dbl>,
## #   disaggregatingdimension2valuecod <dbl>, disaggregatingdimension3 <dbl>,
## #   disaggregatingdimension3valuecod <dbl>, datasourcedimension <dbl>,
## #   datasourcedimensionvaluecode <dbl>, value <chr>, numericvalue <dbl>,
## #   low <dbl>, high <dbl>, comments <dbl>, date <chr>, ...
```

```
tail(itn_access1, 9) # obtain last 9 rows of dataframe
```

```
## # A tibble: 9 x 14
##   IndicatorCode        Indicator     ValueType ParentLocationCode ParentLocation
##   <chr>                <chr>         <chr>     <chr>              <chr>
## 1 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 2 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 3 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 4 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 5 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 6 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 7 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 8 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## 9 MALARIA_ITN_COVERAGE Population w~ numeric   AFR                Africa
## # i 9 more variables: `Location type` <chr>, SpatialDimValueCode <chr>,
## #   `Period type` <chr>, Period <dbl>, IsLatestYear <lgl>,
## #   FactValueNumeric <dbl>, Value <dbl>, Language <chr>, DateModified <chr>
```

QN 3; View the first 10 columns for the first 30 countries in the MALARIA_TOTAL_CASES. dataset

```
#using indexing
malaria_total_cases_df[1:30,1:10]
```

```
## # A tibble: 30 x 10
##         id indicatorcode       spatialdimension spatialdimensionvaluecode
##      <dbl> <chr>               <chr>            <chr>
##  1 6653181 MALARIA_TOTAL_CASES COUNTRY          PAK
##  2 5140438 MALARIA_TOTAL_CASES COUNTRY          MWI
##  3 3504290 MALARIA_TOTAL_CASES COUNTRY          MDG
##  4 8663118 MALARIA_TOTAL_CASES COUNTRY          CAF
##  5 4520145 MALARIA_TOTAL_CASES COUNTRY          AFG
##  6 6939956 MALARIA_TOTAL_CASES COUNTRY          MLI
```

```
##  7 3804728 MALARIA_TOTAL_CASES COUNTRY          AFG
##  8 9255117 MALARIA_TOTAL_CASES COUNTRY          MOZ
##  9 2516740 MALARIA_TOTAL_CASES COUNTRY          ZAF
## 10 2556863 MALARIA_TOTAL_CASES COUNTRY          BRA
## # i 20 more rows
## # i 6 more variables: parentlocationcode <chr>, parentlocation <chr>,
## #   timedimension <chr>, timedim <dbl>, disaggregatingdimension1 <dbl>,
## #   disaggregatingdimension1valuecod <dbl>
```

QN 4; From the "ITN access" sheet of ITN Access.xlsx, display all rows for the following columns: Indicator, ParentLocation, SpatialDimValueCode, Period, and FactValueNumeric

```
#using indexing
itn_access1[1:360, c("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeric"]
```

```
## # A tibble: 360 x 5
##    Indicator          ParentLocation SpatialDimValueCode Period FactValueNumeric
##    <chr>              <chr>          <chr>                <dbl>            <dbl>
##  1 Population with a~ Africa         GAB                   2023             13.0
##  2 Population with a~ Eastern Medit~ DJI                   2023             24.9
##  3 Population with a~ Africa         AGO                   2023             27.0
##  4 Population with a~ Eastern Medit~ SOM                   2023             28.5
##  5 Population with a~ Africa         ETH                   2023             28.5
##  6 Population with a~ Africa         GNQ                   2023             29.0
##  7 Population with a~ Africa         GNB                   2023             33.3
##  8 Population with a~ Africa         MRT                   2023             38.7
##  9 Population with a~ Africa         ERI                   2023             41.7
## 10 Population with a~ Africa         MWI                   2023             47.1
## # i 350 more rows
```

```
#using dplyr
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
itn_access1  %>% select("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeri
```

QN 5; Using the str() function, determine the data type for each of the following columns: Indicator, ParentLocation, SpatialDimValueCode, Period, and FactValueNumeric in the "ITN access" sheet of ITN Access.xlsx.

```
#using indexing
str(itn_access1[c("Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeric")])
```

```
## tibble [360 x 5] (S3: tbl_df/tbl/data.frame)
##  $ Indicator          : chr [1:360] "Population with access to an insecticide-treated bed net (ITN) i
##  $ ParentLocation     : chr [1:360] "Africa" "Eastern Mediterranean" "Africa" "Eastern Mediterranean
##  $ SpatialDimValueCode: chr [1:360] "GAB" "DJI" "AGO" "SOM" ...
##  $ Period             : num [1:360] 2023 2023 2023 2023 2023 ...
##  $ FactValueNumeric   : num [1:360] 13 24.9 26.9 28.5 28.5 ...
```

```
#using dplyr
library(dplyr)
str(select(itn_access1, "Indicator", "ParentLocation", "SpatialDimValueCode", "Period", "FactValueNumeri
```

```
## tibble [360 x 5] (S3: tbl_df/tbl/data.frame)
##  $ Indicator          : chr [1:360] "Population with access to an insecticide-treated bed net (ITN) i
##  $ ParentLocation     : chr [1:360] "Africa" "Eastern Mediterranean" "Africa" "Eastern Mediterranean
##  $ SpatialDimValueCode: chr [1:360] "GAB" "DJI" "AGO" "SOM" ...
##  $ Period             : num [1:360] 2023 2023 2023 2023 2023 ...
##  $ FactValueNumeric   : num [1:360] 13 24.9 26.9 28.5 28.5 ...
```

After using the above methods; the structure of the datasets is as follows; -Indicator is character -ParentLocation is character -SpatialDimValueCode is character -Period is numeric -FactValueNumeric is numeric

QN 6; Using the summary() function, determine the mean, minimum, and maximum number of malaria cases from the Malaria cases, suspected, number.csv dataset.

```
summary(malaria_cases_df) #summarises the Malaria cases, suspected, number.csv dataset.
```

```
##  IndicatorCode       Indicator          ValueType         ParentLocationCode
##  Length:859         Length:859         Length:859         Length:859
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##  ParentLocation     Location.type      SpatialDimValueCode   Location
##  Length:859         Length:859         Length:859          Length:859
##  Class :character   Class :character   Class :character    Class :character
##  Mode  :character   Mode  :character   Mode  :character    Mode  :character
##
##
```

```
##
##  Period.type            Period     IsLatestYear    FactValueNumeric
##  Length:859         Min.   :2015   Mode :logical   Min.   :      121
##  Class :character   1st Qu.:2017   FALSE:758       1st Qu.:   147432
##  Mode  :character   Median :2019   TRUE :101       Median :   651385
##                     Mean   :2019                   Mean   :  4877290
##                     3rd Qu.:2021                   3rd Qu.:  3413658
##                     Max.   :2023                   Max.   :164949610
##     Value              Language         DateModified
##  Length:859         Length:859         Length:859
##  Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
```

```
summary(malaria_cases_df$FactValueNumeric) #focuses on the FactValueNumeric column
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
##       121    147432    651385   4877290   3413658 164949610
```

```
#MALARIA CASES
#mean was 4877290
#minimum was  121
#maximum was 164949610
```

ASSIGNMENT 2

Merge/ bring in the "location" column from itn_access2 to itn_access1

```
#using dplyr
itn_merged <- left_join(itn_access1, itn_access2, by="SpatialDimValueCode")
```

```
## Warning in left_join(itn_access1, itn_access2, by = "SpatialDimValueCode"): Detected an unexpected ma
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 8 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```
str(itn_merged) # using the str() function; you can see the the location column has been added
```

```
## tibble [3,240 x 15] (S3: tbl_df/tbl/data.frame)
##  $ IndicatorCode      : chr [1:3240] "MALARIA_ITN_COVERAGE" "MALARIA_ITN_COVERAGE" "MALARIA_ITN_COVE
##  $ Indicator          : chr [1:3240] "Population with access to an insecticide-treated bed net (ITN)
##  $ ValueType          : chr [1:3240] "numeric" "numeric" "numeric" "numeric" ...
##  $ ParentLocationCode : chr [1:3240] "AFR" "AFR" "AFR" "AFR" ...
##  $ ParentLocation     : chr [1:3240] "Africa" "Africa" "Africa" "Africa" ...
##  $ Location type      : chr [1:3240] "Country" "Country" "Country" "Country" ...
##  $ SpatialDimValueCode: chr [1:3240] "GAB" "GAB" "GAB" "GAB" ...
##  $ Period type        : chr [1:3240] "Year" "Year" "Year" "Year" ...
```

```
##  $ Period             : num [1:3240] 2023 2023 2023 2023 2023 ...
##  $ IsLatestYear       : logi [1:3240] TRUE TRUE TRUE TRUE TRUE TRUE ...
##  $ FactValueNumeric   : num [1:3240] 13 13 13 13 13 ...
##  $ Value              : num [1:3240] 13 13 13 13 13 ...
##  $ Language           : chr [1:3240] "EN" "EN" "EN" "EN" ...
##  $ DateModified       : chr [1:3240] "2024-12-10T05:00:00.000Z" "2024-12-10T05:00:00.000Z" "2024-12-
##  $ Location           : chr [1:3240] "Gabon" "Gabon" "Gabon" "Gabon" ...
```